

D303

Azure Fundamentals

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A retailer with 40 stores moves its inventory app to Azure virtual machines (IaaS) in the East US region. The CIO hopes to cut costs by 20%. Who is responsible for installing operating system security patches on those VMs?

- A. Microsoft
- B. Microsoft, because the provider is always responsible for security in the cloud.
- C. Whoever is in the East US region's datacenter
- D. The retailer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

For which responsibility is the customer accountable in every cloud service model, including SaaS?

- A. Operating system patching
- B. Physical datacenter security
- C. Its data and the accounts and identities that access it
- D. Network hardware maintenance

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A law firm with a downtown office replaces a \$250,000 server purchase with Azure services billed monthly by usage. The managing partner prefers predictable costs. How is the new spending classified?

- A. CapEx, because the amount is large.
- B. Operational expenditure (OpEx)
- C. Capital expenditure (CapEx)
- D. Neither, because cloud bills are not expenses.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A bank headquartered in Chicago must keep customer records in its own datacenter for regulatory reasons, but wants to run seasonal analytics jobs in Azure. The bank has 2,000 employees. Which deployment model fits?

- A. Multi-region public cloud
- B. Private cloud
- C. Public cloud
- D. Hybrid cloud

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A streaming service with 3 million subscribers wants its web app in West Europe to keep running if a single datacenter in that region loses power. The team uses Visual Studio. What should it use?

- A. Deploy across availability zones in West Europe.
- B. Add more subscribers to a second region.
- C. Place the VMs on a single larger server.
- D. Deploy to the paired region only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A payroll app built by a team of six needs both an App Service web app with a 99.95% SLA and an Azure SQL Database with a 99.99% SLA. Both must be up for the app to work. The app processes 8,000 paychecks a month. What is the composite SLA?

- A. 99.95%
- B. 99.99%
- C. About 99.94%
- D. 99.97%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Two independent instances each have 99% availability, and the app works if either one is up. What is the combined availability?

- A.** 98.01%
- B.** 99%
- C.** 100%
- D.** 99.99%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A ticket site owned by a sports league sees traffic jump tenfold for two hours when playoff seats go on sale, then drop back. The site's logo was redesigned last year. The team wants Azure to add instances during the rush and remove them afterward without anyone stepping in. Which capability is this?

- A.** Elasticity
- B.** Vertical scaling
- C.** High availability
- D.** Scaling out only, since removal is never automatic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

An administrator changes a single VM from 4 vCPUs to 16 vCPUs to handle a heavier workload. What kind of scaling is this?

- A.** Horizontal scaling (scaling out)
- B.** Vertical scaling (scaling up)
- C.** Scaling by 4 instances
- D.** Elasticity

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A healthcare company with 12 subscriptions must ensure no one creates resources outside US regions. The compliance officer, who reports to the board, wants the rule enforced automatically. Which Azure feature should be used?

- A. Role-based access control
- B. A resource lock
- C. Azure Policy assigned at a management group above the subscriptions
- D. Twelve separate policies, one per subscription, because policies cannot be inherited.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

An administrator wants to make sure a production database cannot be deleted by accident, even by users with Owner rights. What should be applied?

- A. A CanNotDelete resource lock
- B. Multifactor authentication
- C. A Reader role for all users
- D. An Azure Policy for allowed regions

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A consulting firm with offices on three continents requires staff to enter a code from a phone app after their password when signing in to Azure. The firm's IT team has five people. What does this control add?

- A. Authorization to specific resources
- B. Stronger authentication through multifactor verification
- C. Guaranteed protection from all account compromise
- D. Encryption of stored data

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A retailer ~~with 40 stores~~ moves its inventory app to Azure virtual machines (IaaS) ~~in the East US region. The CIO hopes to cut costs by 20%.~~ Who is responsible for installing operating system security patches on those VMs?

Target word: who patches the OS in IaaS **Decides:** In IaaS the customer manages the OS.

A	Adjacent category	True for PaaS and SaaS, not IaaS.
B	Absolute language	Security is shared; the split depends on the service model.
C	Camouflage	Region location does not change the responsibility split.
D	Correct	IaaS leaves the OS, apps, and data to the customer.

2. Answer C Concept 1 • transfer

For which responsibility is the customer accountable in every cloud service model, including SaaS?

Target word: reverse: task the customer always keeps **Decides:** Data, accounts, and access always stay with the customer.

A	Confusion set	Only in IaaS; the provider handles it in PaaS and SaaS.
B	Inverted operation	Always the provider's.
C	Correct	These never shift to the provider.
D	Adjacent category	Provider-managed infrastructure.

3. Answer B Concept 2

A law firm ~~with a downtown office~~ replaces a \$250,000 server purchase with Azure services billed monthly by usage. ~~The managing partner prefers predictable costs.~~ How is the new spending classified?

Target word: expense type of cloud consumption **Decides:** Pay-as-you-go billing for services.

A	Wrong stem number	Classification depends on the type of spending, not the \$250,000 figure.
B	Correct	Ongoing usage-based service spending is OpEx.
C	Inverted operation	CapEx describes the server purchase being replaced.
D	Absolute language	Cloud services are an operating expense.

4. Answer D Concept 3

A bank ~~headquartered in Chicago~~ must keep customer records in its own datacenter for regulatory reasons, but wants to run seasonal analytics jobs in Azure. ~~The bank has 2,000 employees.~~ Which deployment model fits?

Target word: deployment model **Decides:** Some workloads stay on-premises while others run in public cloud, connected.

A	Confusion set	Still all public; nothing stays in the bank's datacenter.
B	Adjacent category	Drops the Azure analytics workloads.
C	Dropped element	Drops the requirement to keep records on-site.
D	Correct	Combines private on-premises resources with public cloud.

5. Answer A Concept 4

A streaming service with 3 million subscribers wants its web app in West Europe to keep running if a single datacenter in that region loses power. The team uses Visual Studio. What should it use?

Target word: protection against one datacenter failure **Decides:** The failure scope is one datacenter within one region.

A	Correct	Zones are separate datacenters in the same region.
B	Camouflage	Subscriber count does not decide resilience design.
C	Inverted operation	Concentrates risk in one place.
D	Adjacent category	Region pairs are for regional disasters; heavier than needed here.

6. Answer C Concept 5

A payroll app built by a team of six needs both an App Service web app with a 99.95% SLA and an Azure SQL Database with a 99.99% SLA. Both must be up for the app to work. The app processes 8,000 paychecks a month. What is the composite SLA?

Target word: composite SLA for dependent services **Decides:** Both must work, so multiply.

A	Confusion set	Takes the lower single SLA; the chain is weaker than either link.
B	Inverted operation	Takes the best SLA as if dependencies added strength.
C	Correct	$0.9995 \times 0.9999 = 0.9994$, or about 99.94%.
D	Adjacent category	Averages the SLAs instead of multiplying.

7. Answer D Concept 5 · transfer

Two independent instances each have 99% availability, and the app works if either one is up. What is the combined availability?

Target word: availability of redundant parallel copies **Decides:** The app fails only if both fail: $1 - (0.01 \times 0.01)$.

A	Inverted operation	0.99×0.99 is the series formula, for when both must work.
B	Dropped element	Ignores the second instance.
C	Absolute language	Redundancy lowers the chance of outage but never removes it.
D	Correct	$1 - 0.0001 = 0.9999$.

8. Answer A Concept 6

A ticket site owned by a sports league sees traffic jump tenfold for two hours when playoff seats go on sale, then drop back. The site's logo was redesigned last year. The team wants Azure to add instances during the rush and remove them afterward without anyone stepping in. Which capability is this?

Target word: name the capability **Decides:** Automatic scaling out and back in with demand.

A	Correct	Resources adjust automatically to match changing demand.
B	Confusion set	Adds power to one instance rather than adding instances.
C	Adjacent category	About staying up during failures, not matching load.
D	Absolute language	Autoscale rules can remove instances too.

9. Answer B Concept 6 · transfer

An administrator changes a single VM from 4 vCPUs to 16 vCPUs to handle a heavier workload. What kind of scaling is this?

Target word: type of scaling **Decides:** More power added to one existing resource.

A	Inverted operation	Would add more VMs, not a bigger one.
B	Correct	Resizing one VM is vertical.
C	Wrong stem number	No instances were added; 4 was the old vCPU count.
D	Adjacent category	Elasticity is automatic; this is a manual resize.

10. Answer C Concept 7

A healthcare company with 12 subscriptions must ensure no one creates resources outside US regions. The compliance officer, ~~who reports to the board~~, wants the rule enforced automatically. Which Azure feature should be used?

Target word: tool that enforces rules on resources **Decides:** The rule is about resource properties (location).

A	Confusion set	RBAC controls who can act, not which regions are allowed.
B	Adjacent category	Locks prevent deletion or change; they do not restrict location.
C	Correct	Policy enforces allowed locations and inherits down to all subscriptions.
D	Wrong stem number	Policies at a management group are inherited by all 12.

11. Answer A Concept 7 · transfer

An administrator wants to make sure a production database cannot be deleted by accident, even by users with Owner rights. What should be applied?

Target word: tool to prevent deletion **Decides:** Protecting the resource itself from deletion.

A	Correct	Locks block deletion regardless of role until removed.
B	Inverted operation	Proves identity; a signed-in Owner could still delete.
C	Adjacent category	Changes permissions and would also block normal work; Owners could still delete.
D	Confusion set	Controls where resources are created, not deletion.

12. Answer B Concept 8

A consulting firm ~~with offices on three continents~~ requires staff to enter a code from a phone app after their password when signing in to Azure. ~~The firm's IT team has five people.~~ What does this control add?

Target word: what MFA strengthens **Decides:** A second factor at sign-in.

A	Confusion set	Permissions come from RBAC after sign-in.
B	Correct	MFA adds a second proof of identity.
C	Absolute language	MFA greatly reduces risk but does not eliminate it.
D	Adjacent category	MFA does not encrypt data.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Shared responsibility by service model	The customer always owns its data, identities, and access; in IaaS the customer also manages the OS and apps, in PaaS the provider manages the OS, and in SaaS the provider manages nearly everything else.	Thinking the cloud provider patches the OS on an IaaS virtual machine.	A service model named and a task such as patching, data, or physical security.
2	CapEx versus OpEx	Capital expenditure is up-front spending on physical assets; operational expenditure is pay-as-you-go spending on services; public cloud consumption is OpEx.	Calling a cloud subscription a capital expense.	Buying servers versus paying a monthly usage bill.
3	Cloud deployment models	Public cloud runs on provider hardware shared across tenants; private cloud serves one organization; hybrid connects both so workloads can move between them.	Calling any on-premises datacenter a hybrid cloud.	Data that must stay on-site while other workloads burst to Azure.
4	Regions, availability zones and region pairs	Availability zones are separate datacenters within one region and protect against a datacenter failure; region pairs protect against a regional disaster.	Using availability zones to survive the loss of a whole region.	The scope of the failure: a rack, a datacenter, or a region.
5	Composite SLA	For services that all must work, multiply their SLAs, so the composite is lower than any single one; redundant parallel copies raise availability.	Taking the lowest SLA, or averaging them.	Two or more dependent services each with an SLA percentage.
6	Scalability versus elasticity	Scaling up (vertical) adds power to one resource; scaling out (horizontal) adds more instances; elasticity is scaling automatically with demand, both out and in.	Calling a one-time manual upgrade elasticity.	Demand that rises and falls, and whether resources adjust automatically.
7	Governance tools	Azure Policy enforces rules on resource properties; RBAC controls who can do what; resource locks prevent deletion or change; management groups sit above subscriptions, which sit above resource groups.	Using RBAC to restrict which regions resources can be created in.	Whether the rule is about the user or about the resource.

#	Concept	Rule	Trap	Tell
8	Authentication versus authorization	Authentication proves who you are (Microsoft Entra ID, MFA); authorization decides what you can do (RBAC); Conditional Access applies if-then signals at sign-in.	Treating MFA as a permission control.	Proving identity versus granting access to a resource.

Course scope (public description): Azure Fundamentals provides the learner with skills needed to describe the following concepts: cloud concepts; core Azure services; core solutions and management tools on Azure; general security and network security features; identity, governance, privacy, and compliance features; and Azure cost management and Service Level Agreements. Learners will gain foundational knowledge of cloud services and how those services are provided with Microsoft Azure. This course is intended for students who are just beginning to work with cloud-based solutions and services or are new to Azure. Competency in this course is demonstrated by successfully completing the Microsoft Azure Fundamentals certification exam (AZ-900). There are no prerequisites to this course.